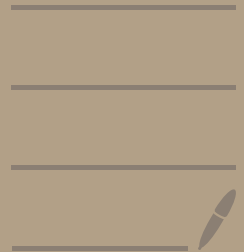


Repaso T4

Tema 15 - Libro Pearson



Problemas Resueltos:

Maquinas Termicas

15.1 -

$$a) e = 1 - \frac{T_c}{T_h} = 1 - \frac{220}{500} = 0,56 = 56\%$$

$$b) e = \frac{W}{Q_h} \rightarrow W = 0,56 \cdot 1500 \cdot 10^3 = 8,4 \cdot 10^5 \text{ J}$$

$$c) W = Q_T \rightarrow W = Q_h - Q_c \rightarrow Q_c = -6,6 \cdot 10^5 \text{ J}$$

15.2 -

Refrigeración

$$T_c = -5^\circ\text{C} = 268,15 \text{ K} \quad T_h = 22^\circ\text{C} = 295,15 \text{ K}$$

$$Q_c = 1,92 \cdot 10^5 \text{ J al día}$$

$$k = \frac{T_c}{T_h - T_c} = \frac{268,15}{295,15 - 268,15} = 9,93$$

$$n = \frac{Q_c}{|W|} \rightarrow |W| = \frac{1,92 \cdot 10^5}{9,93} = 19332,5 \text{ J}$$

$$P = \frac{|W|}{t} \rightarrow P = \frac{19332,5}{24 \cdot 3600} = 0,224 \text{ W}$$

15.3 -

$$a) T_h = 20^\circ\text{C} = 293,15 \text{ K} \quad T_c = 10^\circ\text{C} = 283,15 \text{ K}$$

$$Q_c = 4,8 \cdot 10^8 \text{ J} \quad P = 250 \text{ W}$$

$$P = \frac{W}{t} \rightarrow W = 250 \cdot 3600 \cdot 24 = 2,16 \cdot 10^7 \text{ J}$$

$$n = \frac{Q_c}{W} = 22,22$$

$$b) n_{\text{carnot}} = \frac{283,15}{293,15 - 283,15} = 28,315$$

$28,315 \neq 22,2 \rightarrow$ Ciclo Irreversible

15. u -

$$\delta = 1,4 \rightarrow C_p = \frac{7}{2} R \rightarrow C_v = \frac{5}{2} R \quad T_H = 317 \text{ K}$$

$$P_1 = 400 \cdot 10^3 \text{ Pa} \quad V_1 = 0,67 \cdot 10^{-3} \text{ m}^3 \quad P_2 = 210 \cdot 10^3 \text{ Pa}$$

$$P_3 = 100 \cdot 10^3 \text{ Pa}$$

a) 1 $\rightarrow$ 2:

$$W = \int_{V_1}^{V_2} p(V) dV \rightarrow W = nRT \ln(V_2/V_1)$$

$$P_2 V_2 = P_1 V_1 \rightarrow 210 \cdot 10^3 V_2 = 400 \cdot 10^3 V_1 \rightarrow V_2 = \frac{40}{21} V_1 = 1,28 \cdot 10^{-3} \text{ m}^3$$

$$pV = nRT \rightarrow n = \frac{4 \cdot 10^5 \cdot 0,67 \cdot 10^{-3}}{8,314 \cdot 317} = 0,102 \text{ mol}$$

$$W = 0,102 \cdot 8,314 \cdot 317 \cdot \ln\left(\frac{40}{21}\right) = 172,69 \text{ J}$$

En proceso isotermico: $Q = W \rightarrow Q = 172,69 \text{ J}$

2 $\rightarrow$ 3: $Q = 0 \rightarrow$ Proceso Adiabático

$$W = \frac{1}{\delta-1} (P_2 V_2 - P_3 V_3) \quad P_2 V_2^\delta = P_3 V_3^\delta$$

$$210 \cdot 10^3 \cdot (1,28 \cdot 10^{-3})^{1,4} = 100 \cdot 10^3 \cdot (V_3)^{1,4}$$

$$V_3 = 2,17 \cdot 10^{-3} \text{ m}^3$$

$$W = \frac{1}{\delta-1} (210 \cdot 10^3 \cdot 1,28 \cdot 10^{-3} - 100 \cdot 10^3 \cdot 2,17 \cdot 10^{-3}) = 127,98 \text{ J}$$

3 $\rightarrow$ 4: $Q = W$

$$W = nRT \ln\left(\frac{V_4}{V_3}\right) \quad T_H V_2^{\gamma-1} = T_c V_3^{\gamma-1} \rightarrow T_c = \frac{T_H V_2^{\gamma-1}}{V_3^{\gamma-1}} = 256,4 \text{ K}$$

$$P_1 V_1 = P_2 V_2 \rightarrow P_4 = \frac{1 \cdot 10^5 \cdot 2,17 \cdot 10^{-3}}{V_4}$$

$$P_1 V_1^\delta = P_4 V_4^\delta \rightarrow 4 \cdot 10^5 \cdot 0,67 \cdot 10^{-3} = \frac{1 \cdot 10^5 \cdot 2,17 \cdot 10^{-3}}{V_4} \cdot V_4^\delta$$

$$V_4^{0,4} = \frac{4 \cdot 10^5 \cdot (0,67 \cdot 10^{-3})^{1,4}}{1 \cdot 10^5 \cdot 2,17 \cdot 10^{-3}} \Rightarrow V_4 = 1,14 \cdot 10^{-3} \text{ m}^3$$

$$W = Q = 0,102 \cdot 8,314 \cdot 256,4 \cdot \ln\left(\frac{1,14 \cdot 10^{-3}}{2,17 \cdot 10^{-3}}\right) = -139,70 \text{ J}$$

$$4 \rightarrow 1 : Q = 0$$

$$W = n (C_v (T_1 - T_3)) = 0,102 \cdot \frac{5}{2} \cdot 8,314 \cdot (256,4 - 317) = -127,98 \text{ J}$$

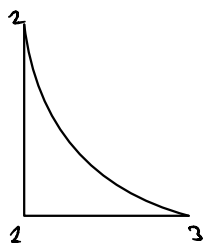
$$W_{23} = -W_{41}$$

$$b) e_{\text{carnot}} = 1 - \frac{T_c}{T_H} = 0,19 = 19\%$$

15.5 -

$$K = \frac{T_c}{T_H - T_c} \rightarrow K = \frac{256,4}{317 - 256,4} = 4,24$$

15.6 -



$$1: (p_1, V_1) \quad 2: (3p_1, V_2) \quad 3: (p_1, 4V_1)$$

$$T_1 = 300 \text{ K} \quad T_2 = 900 \text{ K} = T_3$$

Monatomico, gas Ideal

$$e = \frac{W}{Q_H}$$

1 $\rightarrow$ 2 $\rightarrow$ Isocórico $\rightarrow W = 0$

$$Q = n C_v (T_2 - T_1) = 600 \cdot \frac{3}{2} \cdot 8,314 \cdot n = 7482,6 n \text{ J}$$

$$W = 0$$

2 $\rightarrow$ 3:

$$Q = W = n R T \ln \left(\frac{V_3}{V_2} \right) = 900 \cdot 8,314 \cdot n \cdot \ln \left(\frac{4V_2}{V_2} \right) = 7482,6 \ln(4) n$$

3 $\rightarrow$ 1

$$Q = n C_p (T_1 - T_3) = -600 \cdot \frac{5}{2} \cdot 8,314 n = -12471 n$$

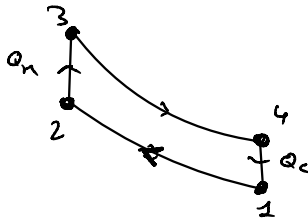
$$W = p_1 (V_2 - V_3) = -3p_1 V_1$$

$$p_1 V_2 = n R T_2 \rightarrow p_1 V_1 = 2494,2 n$$

$$W = 7482,6 n \text{ J}$$

$$e = \frac{7482,6 \ln(4) n - 7482,6 n}{7482,6 n + 7482,6 \ln(4) n} = 0,162 = 16,2\%$$

$$15.7 - \quad \frac{1}{Q_{otto}} = \frac{1}{r^{\gamma-1}} = \frac{W}{Q_H}$$



$$W_{23} = W_{41} = 0$$

$$Q_{34} = Q_{12} = 0$$

$$Q_H = n C_V \Delta T \rightarrow Q_H = n \cdot C_V \cdot (T_3 - T_2)$$

$$W_T = W_{34} + W_{12} = n C_V (T_3 - T_4) + n C_V (T_1 - T_2)$$

$$e_{otto} = \frac{n C_V (T_3 - T_4 + T_1 - T_2)}{n C_V (T_3 - T_2)} = \frac{T_3 - T_2}{T_3 - T_2} - \frac{T_4 + T_1}{T_3 - T_2}$$

$$e_{otto} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

$$T_3 V_3^{\gamma-1} = T_4 V_4^{\gamma-1} \rightarrow T_3 V_2^{\gamma-1} = T_1 V_1^{\gamma-1}$$

$$T_4 = T_3 \left(\frac{V_2}{V_1} \right)^{\gamma-1}$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \rightarrow T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

$$e_{otto} = 1 - \frac{\left(\frac{V_1}{V_2} \right)^{\gamma-1} (T_3 - T_2)}{T_3 - T_2} = 1 - \left(\frac{V_1}{V_2} \right)^{\gamma-1} \quad r = \frac{V_1}{V_2}$$

$$e_{otto} = 1 - r^{-(\gamma-1)} = 1 - \frac{1}{r^{\gamma-1}}$$

$$15.8 - \quad r = 8 \quad \gamma = 1.4$$

$$e_{otto} = 1 - \frac{1}{r^{\gamma-1}} = 1 - \frac{1}{8^{1.4-1}} = 0.56 = 56\%$$

Variación de entropía

$$15.9 - \Delta S = \frac{dQ}{T} = \frac{m L_f}{273,15} = \frac{0,5 \cdot 3134 \cdot 10^5}{273,15} = 611,395/K$$

$$15.10. \Delta S = \int \frac{dQ}{T} = \int \frac{mc dT}{T} = mc \ln\left(\frac{T_2}{T_1}\right)$$

$$a) \Delta S = 2,6 \cdot 890 \cdot \ln\left(\frac{290}{650}\right) = -818,393/K$$

$$b) \Delta S = \frac{Q_{rev}}{T} = \frac{2,6 \cdot 390 \cdot (650 - 290)}{290} = 1258,765/K$$

$$c) \Delta S_{uni} = \sum \Delta S = 1258,76 - 818,39 = 440,37 J/K$$

15.11 -

$$m_{20} = 0,05 kg \quad T_{20} = 293,15 K$$

$$m_{80} = 0,04 kg \quad T_{80} = 353,15 K$$

$$a) \Delta Q = 0$$

$$- m_{80} c_{80} \Delta T_{80} + m_{20} c_{20} \Delta T_{20} = 0$$

$$- 0,04 \cdot 4190 \cdot (80 - T) + 0,05 \cdot 4190 \cdot (T - 20) = 0$$

$$+ 167,6 T - 13408 + 209,5 T - 4190 = 0$$

$$377,1 T = 17598 \rightarrow T = 46,7^\circ C = 319,82 K$$

$$b) \Delta S = mc \ln\left(\frac{T_2}{T_1}\right)$$

$$\Delta S_{80} = 0,04 \cdot 4190 \cdot \ln\left(\frac{319,82}{353,15}\right) = -16,65/K$$

$$\Delta S_{20} = 0,05 \cdot 4190 \cdot \ln\left(\frac{319,82}{293,15}\right) = 18,24 J/K$$

$$\Delta S_r = \Delta S_{80} + \Delta S_{20} = 18,24 - 16,6 = 1,62 J/K$$

$$15.12 - \Delta S_{\text{total}} = -18,24 + 16,6 = -1,62 \text{ J/K}$$

$$\Delta S_{\text{fuente}} = 1,62 \text{ J/K}$$

15.13 -

a) $\Delta U = 0 \rightarrow$ Proceso Aislado

b) $n = 0,5 \text{ mol}$

Debemos Idear un proceso isotermico y reversible

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right) = nR \ln\left(\frac{P_1}{P_2}\right)$$

$$V_2 = 2V_1$$

$$\Delta S = nR \ln(2) = 0,5 \cdot 8,314 \cdot \ln(2) = 2,88 \text{ J/K}$$

15.14 -

Proceso Isotermico y suponemos revers. ile.

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right) = nR \ln(2) \quad V_2 = 2V_1$$

$$\Delta S_{\text{He}} = \Delta S_{\text{Ne}} = 0,02 \cdot 8,314 \cdot \ln(2) = 0,115 \text{ J/K}$$

$$\Delta S = \Delta S_{\text{He}} + \Delta S_{\text{Ne}} = 0,23 \text{ J/K}$$

15.15 -

$$dQ = nC_V dT \quad \Delta S = \int \frac{nC_V dT}{T} = nC_V \ln\left(\frac{T_2}{T_1}\right)$$

$$C_V = \frac{5}{2}R \quad \textcircled{1} \text{ A } V \text{ cte}$$

$$\textcircled{2} \text{ A } T \text{ cte}$$

$$\Delta S_1 = \frac{5}{2} \cdot 8,314 \cdot \ln\left(\frac{373,15}{273,15}\right) = 6,49 \text{ J/K}$$

$$\Delta S_2 = 8,314 \cdot \ln\left(\frac{5,65}{1}\right) = 14,4 \text{ J/K}$$

$$\Delta S_{\text{total}} = S_1 + S_2 = 20,85 \text{ J/K}$$

